



ITS65504 Computer and Network Security

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Name (Block Capital)	Student ID	Signature	Marks (For Lecturer Use)
1. LIM KAI ERN	354798	<i>kaiern</i>	
2. WONG LIEN	374089	<i>LIEN</i>	
3. TIAN XINYU	373276	<i>Xinyu</i>	

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ITS65504 Computer and Network Security

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Tasks		LIM KAI ERN	WONG LIEN	TIAN XINYU
1.	Introduction		√	
2.	Ransomware Risks and Impact			√
3.	X.800, NIST Analysis			√
4.	Solutions and Approach	√		
5.	Port Scanning	√		
6.	Conclusion		√	
7.	Video		√	

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Abstract

Living in the technological era, ransomware attacks have become one of the biggest challenges to the cybersecurity of higher learning institutions around the world. According to the research, estimated damages worldwide by 2031 will surpass roughly USD 250 billion (Cybersecurity Ventures, 2023). In fact, ransomware usually does not target universities alone, any enterprise, business such as hospitals, financial institutions, manufacturing companies, and government agencies can be the victim due to their reliance on digital systems and the value of the data they hold. However, Taylor's University in Malaysia, one of the largest private university in the country with more than 30,000 students, fits into this landscape as well and represents a quite high-value target for cyber-attackers because of its scale, diverse user base, and extensive digital services.

This report provides an assessment of the ransomware risk to Taylor's University based on the X.800 Security Architecture Framework and the NIST Cybersecurity Framework. There might a possibility of existing critical vulnerabilities in the measures for the security of endpoints in the university, where the risk may compromise the functionality of the myTIMeS support system, the Student Information System in the university, the examination systems in place, and the financial systems.

Also, the port scanning analysis activities will be included in this report, amongst with analysis an of the MITRE ATT&CK framework to identify specific technical attack vectors. Additionally, the defensive measures presented in this report are structured using the People, Process and Technology (PPT) framework and are derived through a practical 8-step risk management approach, which could ensure that Taylor's University adopts a comprehensive, evidence-based and also systematically developed cybersecurity strategy.

1.0 Introduction

1.1 Overview of Ransomware Threats

Ransomware is malicious software that encrypts victim data and systems to extort ransom payments. Over the past decade, ransomware has evolved from sporadic attacks targeting individuals to highly organized criminal operations against enterprises and institutions. Modern ransomware employs "double-extortion" tactics, where attackers not only encrypt data but also exfiltrate and threaten to publicly disclose sensitive information, significantly increasing pressure on victims to pay (Kaspersky Labs, 2023).

Global ransomware damage costs are projected to surge from approximately USD 20 billion in 2021 to USD 250 billion by 2031 (Cybersecurity Ventures, 2023). CyberSecurity Malaysia reports indicate a significant increase in local ransomware incidents, with educational institutions becoming particularly attractive targets due to their open networks, diverse user populations, and valuable data assets (FBI Cyber Division, 2023). Notably, in 2022, Lincoln College in the United States was forced to close permanently after 157 years of operation following a ransomware attack that rendered critical systems unrecoverable, clearly demonstrating that ransomware can destroy an institution (Mandiant, 2023).

1.2 Why Universities are High-Risk Targets

Learning institutions have some characteristics that make them more susceptible to these attacks. They maintain open networks to support learning activities, and these networks have many entry points for attackers. As stated in the introduction, Taylor's University has more than 30,000 students, which every single student has different levels of awareness concerning information technology. Consequently, the constant turnover of students makes it extremely difficult for the university to maintain a consistent approach to information security (Sophos, 2023).

Besides that, universities keep huge amounts of sensitive information like identification numbers, passports, financial information, educational transcripts, and research information that might be highly valuable on the dark web. At the same time, educational institutions use complex information systems that involve legacy systems, cloud services, third-party services, and a Bring-Your-Own-Device approach. These factors lead to inconsistent security levels across their systems (National Institute of Standards and Technology, 2020). Also, educational institutions follow tightly scheduled timelines. Hence, failures during critical phases such as examinations or admission processes may cause ripple effects, putting the institutions in a

position where even paying ransoms may be seen as a way to quickly restore access to services (Ponemon Institute, 2023).

1.3 Purpose and Scope of This Report

To be more specific, this report provides Taylor's University management with strategic guidance for addressing the dynamic ransomware threat landscape. It includes comprehensive analysis of Taylor's University's ransomware threat, understanding what it entails when a ransomware attack succeeds, what types of operational, financial, reputational and legal threats the organization faces, analyzing the existing cybersecurity posture and the current state of cybersecurity practices, utilizing the X.800 and NIST security guidelines, identifying technical vulnerabilities using a port scanning tool, mapping attacks to the MITRE ATT&CK attack framework, and establishing best practices related to People, Process, and Technology aspects (International Telecommunication Union, 2023; National Institute of Standards & Technology, 2018).

The study scope is about completing the assessment of the existing level of exposure based on the external attack surface, analyzing the existing security controls and incident response preparedness, examining the existing data protection solution set, evaluating the existing security awareness training agenda, and reviewing the existing third party risk management. All the above-point recommendations are done considering the characteristics of Taylor's University, such as its size, technology, PDPA requirements that the organization has to comply with, the academic calendar, and consideration of the costs incurred. Therefore, by addressing these elements, Taylor's University will have better capability to enhance their cybersecurity resilience, secure their data and reputation, ensure business performance, and maintain its stakeholders' confidence (Verizon, 2023; Ransomware Task Force, 2021).

2.0 Ransomware Risks and Impact

2.1 Real-World Ransomware Incidents in Higher Education

Ransomware has become one of the most disruptive threats that affects universities around the world. A very important example happened in 2022 when Lincoln College in the United States was attacked (FBI Cyber Division, 2023). The attack caused long outages in their learning systems, finance systems, and even their recruitment systems. The college struggled to restore their critical services because their backup systems weren't strong enough and they didn't fully understand how much damage had been done. After 157 years of operation, the college eventually decided to close permanently—which is a clear sign that ransomware doesn't just cause temporary problems, it can actually destroy a university (Mandiant, 2023).

Looking at industry reports shows how serious this problem is getting. Based on Cybersecurity Ventures estimates, ransomware damages expected worldwide each year can potentially go beyond USD 250 billion in the year 2031 (Cybersecurity Ventures, 2023). In addition, Mandiant and FBI reports have identified that higher education institutions like Taylor's are particularly vulnerable to ransomware attacks because of the open network, variety of users, and interdependent systems in place (FBI Cyber Division, 2023). In Taylor's case, it has several types of systems that are required to be operating at all times in order to cater for tens of thousands of students and members of the administration.

2.2 Likely Attack Vectors for Taylor's University

Taylor's is a top private university in Malaysia with a lot of digital services, and so it was subject to a number of the common Taylor's faces several common attack methods that ransomware groups like to use:

2.2.1 Phishing and Credential Harvesting

Phishing is probably the most common way that attackers get into universities. They send emails that look like they come from IT support, or from lecturers, or from the admin office. They try to trick people into giving away their login details. Since Taylor's has thousands of accounts on myTIMeS, SIS, Office 365, and other online systems, just one person giving away their password can let attackers get into the whole network and cloud services (Amos, 2019).

2.2.2 Compromised or Unsecured Endpoints

Endpoints are entities such as staff laptops, lab computers, student personal computers, and devices you use when working from home. These devices are vulnerable, because they may be missing critical security updates, may not be configured securely, or may be lacking antivirus software. In the event that an infected device connects, it can then spread the ransomware to other machines within the campus network (Sophos, 2023).

2.2.3 Third-Party Integrations and Vendor Systems

Universities have to use systems from other vendors – for their payment systems, their library platforms, their research tools, their academic software. The issue is that a few of these vendor systems may not be as secure as Taylor's. This means attackers might be able to get into Taylor's through these external vendors instead of attacking Taylor's directly. This kind of supply-chain attack has been a big problem for universities (Cisco Talos, 2023).

2.2.4 Vulnerable Public-Facing Services

Some of Taylor's systems are visible on the internet, like web portals, research servers, and APIs that outside people can access. If these systems don't get security updates quickly enough, or if they have plugins that are out of date, or if they're set up wrong, attackers can find the weak spots and use them to get into the main systems (National Institute of Standards and Technology, 2020).

Since Taylor's has valuable student personal data, financial information, and research work, it's an attractive target for criminal groups that make money by stealing data and demanding ransom payments.

2.2.5 Operational Disruption

If a ransomware attack had successful, immediate widespread disruption would occur. One of them would likely be LMS (myTImeS) downtime that led to stop all online classes. Students couldn't access their learning materials, submit their assignments, or communicate with their lecturers. Furthermore, examination system outages would delay tests, proctoring, grading, and affect when students can progress to the next year or graduate. Additionally, Student Information System (SIS) disruption would stop registration, transcript production, timetable management, and all the academic office work. Financial system interruption would mess up tuition billing, scholarship payments, vendor payments, and employee salaries. Universities have to follow strict academic calendars and timelines, so even short outages cause big

problems, where lots of extra work piles up, staff gets very busy, and administrative risk goes up. If the systems are down for a long time, it could affect the quality of teaching, students might not be happy, and it could break contracts with partners (Verizon, 2023).

2.3 Reputation Damage

The strong reputation of Taylor's University is one of its greatest assets, particularly in terms of attracting students from abroad and collaborating with industry. There would be a public ransomware attack, and that would probably damage how students and parents, companies, and the media think of the university. A few of these are (Ransomware Task Force, 2021):

Students and their parents may fear that Taylor's has been unable to protect their personal data, so they will go to other universities. Firms may hesitate to do research with Taylor's, give it internships or collaborate on projects until the situation seems more robust or secure/ News outlets would be on this story, all negative, which might hurt the university's rankings and the way students and the public perceive it. It generally takes years for universities to regain their credibility and clean up their image after a large security incident. For Taylor's, which attracts students from across the globe, a loss of reputation would bring academic and financial troubles.

2.4 Data Confidentiality Loss

Current ransomware operators now employ a form of "double-extortion" (Kaspersky Labs, 2023). What that entails is that they first steal sensitive information, then they encrypt the systems and finally they ransom by threatening to dump the information online if they don't get a ransom. The following data is at risk at Taylor's, where student personal information including IC numbers, passport numbers, home addresses, email addresses, and academic records Staff information such as payroll details, HR files and personal information. Research, patents and academic papers that has yet to be published Financial statements and other internal documents of administrators Such stolen information can be sold on the dark web, shared publicly on the web, or used for identity theft. It may also trigger issues under Malaysia's Personal Data Protection Act (PDPA), potentially entailing fines or other legal ramifications for the university (Verizon, 2023).

2.5 Financial Loss

The money cost of a ransomware attack goes way beyond just paying the ransom demand.

There are many different costs that add up (Ponemon Institute, 2023):

- Paying experts to investigate what happened and figure out forensics
- Rebuilding and restoring all the systems and computers that were damaged
- Recovering data, checking it's correct, and making sure it's not corrupted
- Hiring lawyers, compliance people, and auditors to help deal with the problem
- Communicating with students, staff, and the media about what happened and making sure everyone affected knows
- Lost productivity when systems are down and people can't do their jobs

For a big university like Taylor's, all these costs can add up to several million ringgit depending on how bad the attack is and how long it takes to get everything working again. After the immediate crisis is solved, the university would also need to spend more money on better security systems to make sure this doesn't happen again.

3.0 X.800 and NIST Analysis

3.1 X.800 Analysis

The evaluation of Taylor's University based on the X.800 security model shows a need for Taylor's University to improve several basic security services to mitigate risks to ransomware (International Telecommunication Union, 2023). In relation to the confidentiality requirement, the partial lack of encryption implemented through backups, on endpoint devices, or in specific database environments exposes data to theft and unauthorized access before encryption of the system takes place. Because of the value of student records, financial data and research information, the uneven use of encryption-at-rest and limited data loss prevention capabilities pose substantial institutional risk. Full end-to-end encryption, stronger access control enforcement, and better cryptographic key management are needed to maintain the security of sensitive information at each point of the data lifecycle.

For integrity, ransomware also inflicts institutional risk, as it allows the unauthorized alteration of academic records, financial records and essential systems (Ponemon Institute, 2023). The university's existing logging infrastructure offers limited visibility and immutability, and it can be tampered with if adversaries successfully escalate privileges. Integrity attacks could go unnoticed and consequently have debilitating effect on operations and reputation if there are no effective audit trails and systematic verification mechanisms in place. Using cryptographic integrity checking, tamper resistant audit logging, and digitally signing for "high value" actions in the system, are recommendations on improving integrity assurance.

Authorization is the third major X.800 requirement, and the practice of Taylor's University is that it is moving but it has not reached the completion. Several core systems, such as Multi-Factor Authentication (MFA), have been implemented, however weaknesses in university-wide MFA coverage and password policy enforcement across the university undermine the strength of the university's authentication. Credential theft continues to be one of the leading vectors for ransomware initial access – and the fact that MFA is not fully adopted leaves an attack window. Improving the security of authentication will be tantalizing need for a very broad MFA deployment on all systems handling sensitive information, tougher password policies, and more robust privileged account management built upon zero-trust verification.

3.2 NIST Analysis

With the NIST Framework for Cybersecurity in mind, there appear to be a number of regions of enhancement required in the Identify, Protect, Detect, Respond, and Recover activities (National Institute of Standards & Technology, 2018). With reference to the Identify activity, the regions of enhancement relate to the lack of asset inventories performed, the need to document legacy environments, and the lack of vulnerability scans. The lack of complete visibility into the environments may leave unknown risks to the environments. There should be a central point for managing environments to facilitate the regions of enhancement.

The “Protect” function points to problems in preventive controls. Though Taylor’s University has mechanisms for backup storage, end-point protection, and firewalls, the lack of immutable storage for backup files, incomprehensive end-point security, particularly for students, and the lack of network segmentation makes it easier for spread of the ransomware attack when the initial point of compromise is breached (Sophos, 2023). Improving the preventive measures would involve the need for isolated and immutable backup designs, improved end-point detection and response systems, and network segmentation for critical operational networks for activities like education, finance, or scientific research.

Within the context of the Detect functionality, the gaps in detection that hinder the university from recognizing the malicious activities at an early stage include the lack of central SIEM system integration or intrusion detection systems. As a result of the fragmented nature of the current logging mechanisms in place, the indicators of possible threats may go undiscovered until considerable damage has been caused (Mandiant, 2023). Implementation of the central logging capabilities, intrusion detection solutions, and the application of behavior analytics would be recommended.

There exists a lack of scenario-driven response playbooks for incidents of a particular type, including ransomware attacks. According to the evaluation of the Respond function assessment tool, the university has high-level processes but a lack of scenario-driven playbooks for incidents of a particular type. To conform to NIST guidelines, the university needs to plan for the processes to be followed in the event of a disruption. Ransomware simulation exercises need to be conducted.

Finally, the Recover function highlights the reliance on the backup and restore processes that have not been thoroughly validated. The RTOs/RPOs for business-critical applications have not been formally established (Verizon, 2023). Improving the restore capabilities would involve

having a plan for the different applications to restore in the event of a disaster. Validation of the applications to restore would have to be established.

4.0 Approaches and Solutions

In general, security approaches toward cyber threats are critical in every organization that it is an undeniable fact, especially living in the technological era. According to CEE Multi-Country News Center (2024), it reveals that there were 600 million cyberattacks happening per day that faced by Microsoft's customers, from both cybercriminals and nation-state actors. Can you even imagine how massive is this cyberattack happened just in a day? Not only this, Microsoft also reported that compared to the previous, there is a 2.75 times increase in ransomware attacks, although the percentage of organizations that are being really ransomed has decreased more than threefold over the past two years. In recent years, ransomware is rising globally, in a new way of payment fraud and misusing legitimate services for phishing and malicious activities. Hence, from this case, and not only this, we can say that it is critical for every organization sets up a prevention to safeguard their network, which we will discuss such preventive strategies in the context of Taylor's University in this section.

Based on the risks identified earlier and guided by international best practices including X.800 security services and the NIST Cybersecurity Framework, this section will cover a comprehensive set of approaches and solutions which could fit in for Taylor's University. Our main goal is to reduce exposure, improve early detection, and ensure the university is able to respond as soon as possible and recover quickly if an incident occurs. Similar to any other cyberthreats, ransomware prevention requires a holistic cybersecurity strategy that strengthens human awareness, institutional processes, and the technological environment. Commonly, technology is always viewed as the first defense in mitigating cyberthreats like firewalls, Security Info and Event Management (SIEM), Intrusion Detection/ Prevention Systems (IDS/ IPS) and more. Nevertheless, cybersecurity is actually not just a technical issue where it involves people, processes, and technology which can referred to as PPT (Eko Yon Handri et al., 2023). Therefore, this project adopts the PPT model as a fundamental structure for protecting IT security systems. Yet, to ensure the PPT recommendations are not generic, theoretical but grounded in a logical and repeatable method, this project incorporates a structured risk-management methodology.

Based on the work of Javaid & Iqbal (2017), we refined and adapted their ideas to form a simplified 8-step ransomware risk-management model tailored specifically for Taylor’s University. We believe this model would help Taylor’s in ensuring ransomware protection is strengthened not only through technical controls but also supported by human behavior and organizational governance. The following subsection explains this 8-step solutions in detail whilst the PPT-based solutions presented later in this chapter are derived directly from this structured approach.

4.1 8-step Practical Ransomware Risk-Management Approach

As stated, this 8-step ransomware risk-management model is based on the reference work of PPT Risk Management Application Model provided by Javaid & Iqbal (2017), where we refined it into more detailed model that is suitable for Taylor’s University. Overall, this model provides a practical and repeatable structure for identifying, assessing, and treating ransomware risks where each step builds on the previous one to ensure that recommendations are not made in isolation but grounded in actual risk analysis and operational needs. Figure1 indicates an overall of the approach.

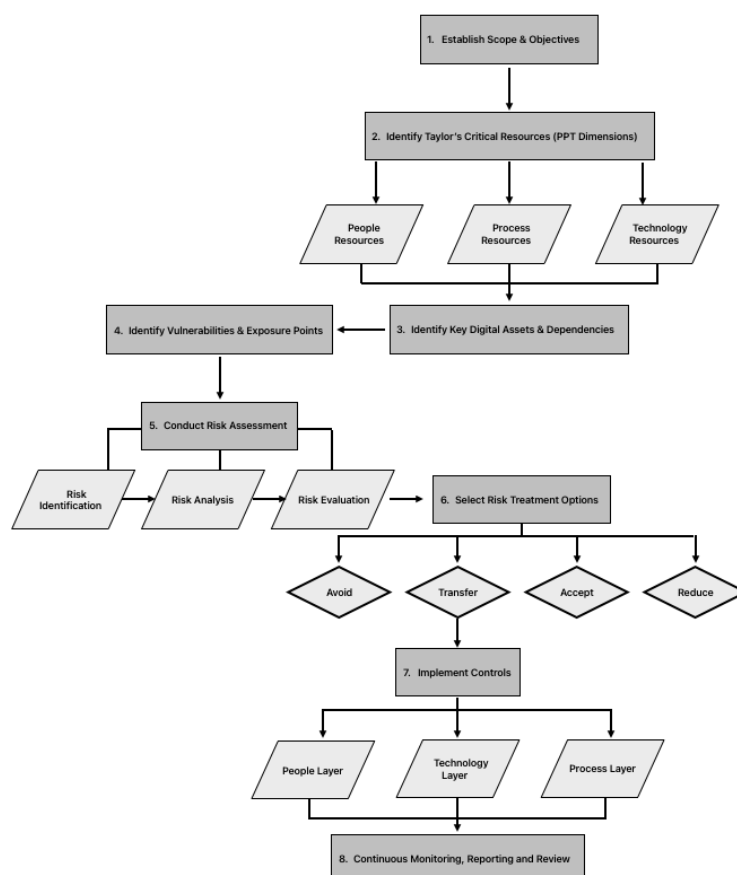


Figure 18-Step Practical Ransomware Risk-Management Approach

4.1.1 Define Scope and Objectives

First of all, the enterprise should establish the scope of ransomware protection efforts by identifying which departments, systems, and digital assets fall under this assessment. This includes core academic systems such as Student Information Systems (SIS), Learning Management System (LMS) which is myTIMeS, financial and HR platforms, faculty and staff endpoints, public-facing web services such as main website (taylors.edu.my), ICT access, resources like e-library, and other essential network infrastructure. So, what are the purpose of these clear objectives? From the work of Javaid & Iqbal (2017), it stated that establishing scope is the first and foremost step to be taken as it will become feasible to correct the issues in later stages. In this case, to be more specific, clear objectives are set to protect sensitive personal data, maintain continuous academic services, support compliance requirements, and minimize operational disruption. On the other hand, defining scope also could help with ensuring that responsibilities, system boundaries, and expected outcomes are understood before moving toward risk analysis or solution design. Hence, this stage involves defining goals and objectives where we mentioned earlier.

4.1.2 Identify Asset Inventory

Coming to the next step where involves creating an inventory of all assets associated with Taylor's University operations across people, processes, and technology. The process includes identifying users who interact with critical systems, academic and administrative workflows that rely on digital platforms, and also which servers, endpoints, and cloud applications are used daily. These assets should be documented with its purpose, owner, location, and type of data handled clearly as the inventory allows the university to understand the full landscape of resources that could be affected by ransomware. Also, it could provide the foundation of evaluating the values and importance of each assets.

4.1.3 Classify Assets by Criticality and Impact

After identifying the assets, they are now be classified based on how essential they are to the university's operations. For instance, systems like SIS, myTIMeS, authentication services, and financial management tools are considered crucial because if downtime happened, they would likely be severely disrupted, and academic or administrative function could be suspended, stopped working. Besides this, staff devices, email systems, and research servers may be essential, while secondary resources like non-critical webpages fall into supporting categories. This procedure helps the enterprise to prioritize where security resources and

protective measures should be focused, especially during high-risk periods such as examinations, enrollment or orientation weeks.

4.1.4 Identify Internal and External Vulnerabilities

This step emphasizes discovering the weaknesses that could enable ransomware to enter or spread within the university. To be precise, external vulnerabilities may include exposed open ports, outdated web services, or insecure configurations discovered through non-intrusive scanning. On the other hand, internal weaknesses often relate to human conduction, like phishing attack, weak passwords, inconsistent patching, or over-reliance on third-party vendors. Furthermore, technology gaps such as outdated software, unmonitored endpoints, or limited segmentation between networks can also contribute to the risk of ransomware. Similar to the previous steps, identifying these vulnerabilities could help Taylor's University understand which and where is most exposed and what kind of improvements are needed urgently.

4.1.5 Risk Assessment

Generally, we can know that not every weaknesses poses the same danger. So, coming to the next step, the university need to evaluate the likelihood of every vulnerability being exposed. Also, predicting of how it would have impact on academic, financial, and operational function is needed as well. For example, phishing may have a high probability due to massive number of users, an unpatched server, on the other hand, could have a lower likelihood but a high impact if compromised. In short, this assessment process transforms discovered weaknesses into a ranked list of risks, which allows Taylor's to focus on the issues that would pose much threat. We believe a structured risk matrix could help in guiding decision-making for next step.

4.1.6 Select Treatment for Different Risk

During this step, Taylor's should decide how each risk should be addressed after prioritizing them. For instance, some risks like unauthorized access, data breaches that have commonly happened these days, can be reduced through stronger controls such as MFA which Taylor has already integrated, segmentation, or enhanced monitoring, while other risks could be avoided by disabling unused services or changing the procedures. Moreover, certain risks also may be transferred via insurance or vendor contracts, whilst low-impact risks might be accepted if the cost of mitigation outweighs the benefit. Briefly, this step discusses about choosing realistic, practical, and cost-appropriate actions that can reduce the chance of ransomware causing major disruption.

4.1.7 Implement Controls

Even though security measures are implemented, it is still not enough to protect the systems. Instead, they must be tested regularly to ensure they are really perform as expected. Hence, this process includes restoring backups to confirm they can actually be used if incident happened, performing phishing simulations to monitor user awareness, carrying out incident-response exercises, verifying if the patches have been applied properly, and tuning detection rules within SIEM or EDR platforms. We believe these validation activities would help Taylor's University identify the gaps in its defenses and refine controls before a real, actual ransomware incident occurs.

4.1.8 Continuous Monitoring, Reporting and Review

In real world, we know that cyber threats evolve constantly, where its escalation endangers innovation, business investment, and the stability of economic, stressing the emergency, critical need for improved cybersecurity (Moore, 2024). Therefore, the final step of this approach involves ongoing monitoring and continuous improvement. The most common and effective way is to perform monthly vulnerability scans. Furthermore, reviewing firewall logs, updating policies, tracking new ransomware techniques are required to ensure cyberthreats are being blocked, especially new, evolved threats. Also, reassessing residual risks that could not be completely eliminated is critical as well. This strategy of monitoring continuously could make sure that the university remains protected as technology, threats, and academic needs change. More than this is it allows risk owners to adapt strategies, ensuring long-term resilience rather than a one-time compliance exercise.

4.2 PPT-Based Solution

After completing the 8-step ransomware risk-management model, we believe that the main vulnerabilities, priorities, and handling options for Taylor's University become much clearer. This model not only underlines where the university is most vulnerable, but also reveals which areas, people, process, or technology need the most attention. In order to translate the results of that structured assessment into even more realistic, practical action, this section presents the PPT-based solutions. As we mentioned at section 5.0, this PPT-based solutions would be derived from the previous 8-step structured approach, where each component of it addresses a different dimension of ransomware resilience including human behavior, institutional workflows, and the technical infrastructure that supports daily operations.

Together, our team believe they could offer a balanced and actionable strategy for strengthening Taylor's University's defenses.

4.2.1 People Layer

From the people aspect, this dimension focuses on the individuals who involved in the process. The phrase "Right time, right time, right people equal success" reflects the Human dimension of the Process Improvement Model, where it highlights the importance of equipping people with the right knowledge, skills, and motivation, whilst encouraging them to take part in continuous improvement. Ultimately, people are the most valuable asset of a company possesses (Prodan et al., 2015). In this case, the people component focuses on the human aspects of Taylor's University's cybersecurity posture. As the previous assessment showed, we already know that human behavior such as clicking on phishing emails, using weak passwords, unsafe downloads, or failing to report suspicious activity, remains one of the most common entry points for ransomware. Hence, it is undeniable that human layer is the university's largest exposure point, specially as an open enterprise.

To strengthen this layer, we suggested that the university may implement various ongoing cybersecurity awareness programmes that educate the users, students and staff on identifying phishing attempts, safe browsing habits, and responsible data handling. This is a fundamental procedure as not every person is aware of the cyber threats, and one's mistake could lead to the university's system failure. Moreover, practical simulations like controlled phishing tests, could be practiced as it can help with measuring improvement over time and identify groups that may require additional, supporting guidance. It is a good thing that Taylor's University integrated multi-factor authentication as stronger authentication practices are important as well. Some other practices including encouraging password managers, enforcing strong password policies could be integrated to even more reinforce, assist in reducing the risk of compromised accounts. Lastly, establishing clear and easily accessible reporting channels can allow user to report suspicious activities quickly, which enabling early detection and faster incident response.

4.2.2 Process Layer

In here, the process layer is about addressing the workflows, procedures, and governance mechanisms which determine how cybersecurity is managed across the enterprise. Strong processes act as the backbone of any ransomware resilience strategy and based on the findings from the risk-management model, we think there are several process areas require

attention. First and foremost, Taylor's should maintain a clearly defined Incident Response Plan (IRP) which emphasizes on immediate actions, communication steps, and recovery procedures in the event of ransomware attack. For the information, an IRP is basically a written document that legally approved by the senior leadership team which helps the organization before, during, and after a confirmed or suspected security incident (Cichonski et al., 2012). Hence, regular tabletop exercises could ensure that employees are understand and aware of their roles and able to respond confidently under pressure. Furthermore, backup and recovery processes also need to be consistent and reliable. For instance, Taylor's could use the 3-2-1 backup strategy, which is a robust data protection plan that keep three copies of the data on two different media with one copy offsite, and critically combined with regular restoration testing to ensure those backups are actually work when needed. This strategy could ensure that important academic and administrative data able to be recovered without paying ransom when incident happened.

Furthermore, the patch management ought to follow a predictable and well-documented cycle so that vulnerabilities in operating systems and applications are addressed in a promptly manner. In addition, third-party vendors who integrate with the university's system should be evaluated for security compliance to prevent supply-chain weaknesses. In short, strengthening these processes improves both readiness and operational continuity.

4.2.3 Technology Layer

The last layer, technology layer is prompt to encompass the infrastructure and security tools that protect Taylor's University's digital environment. We believe most of the enterprise these days, especially former, old ones, their implemented technology that used to mitigate cyberattack still pose inherent risks due to new or outdated technologies, yet in this case, several technical risks were identified in the previous steps, including exposed public-facing services, inconsistent patching, limited network segmentation, and a reliance on basic malware defenses. Accordingly, to mitigate effectively these issues, Taylor's should continue expanding multi-factor authentication across all major systems, ensuring that compromised passwords alone are unable to grant attackers access. Additionally, advanced endpoint protection can be integrated, such as Endpoint Detection and Response (EDR), which provides early detection of suspicious behavior, including file encryption or unauthorized access attempts.

Again, network segmentation is crucial as well for limiting the spread of ransomware which student networks, administrative systems, and research environments should operate on independent segments in order to reduce potential impact. Moreover, email security gateways with attachment sandboxing can help in blocking malicious files before they reach inboxes where the university may use this approach. Finally, we believe maintaining offline or immutable backups remains the ultimate safeguard against ransomware encryption, ensuring that operations can be stored efficiently even if prevention measures somehow fail.

5.0 Port Scanning Results and Analysis

In the cybersecurity field, it is a common knowledge that port scanning is a crucial part of assessing ransomware exposure, since accessible services could be exploited by attackers to gain initial access into an organisation. Therefore, for this project, a non-intrusive, external-only port scanning activity was conducted on Taylor's University's public -facing domains. This could ensure that while the assessment remains ethical which does not interfere with university operations, it also aligns with the requirements to test only what is openly accessible on the internet.

5.1 Scope and Methodology of Scanning

The external port scanning was conducted on 4 publicly accessible Taylor's University systems using Nmap. Again, only public-facing and non-restricted systems were scanned, which any users on the internet can access and therefore pose potential exposure risks. All scans were executed using the regular TCP, UDP scan and service detection like `nmap`, `nmap -sV`, `-sU` where were taught in the Port Scanning Fundamentals lecture materials.

The following domains were included in the port scanning activity:

- `university.taylors.edu.my`
- `mytimes.taylors.edu.my`
- `campuscentral.taylors.edu.my`
- `taylorslibrary.taylors.edu.my`

5.2 Results of Scanning

Based on the performing, we observed that a same consistent pattern had resulted across all domains where only port 80 (HTTP) and port 433 (HTTPS) were open. This situation is normal since these ports are expected for any publicly accessible web service, yet do not indicate a security weakness by themselves. On the other sides, service detection using `-sV` returned `tcpwrapped` for both ports, this circumstance has also commonly happened when server is protected behind a cloud load balancer such as AWS, Microsoft Azure, or an enterprise wen application firewall. Hence, this results indicate that the underlying service information is intentionally hidden, which could reduce fingerprinting opportunities for the attackers. The following figure (fig. 1) shows the example of results using the domain of `mytimes.taylors.edu.my`.

```

(kaiern@carrot)-[~]
$ nmap mytimes.taylors.edu.my
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-08 10:20 +08
Nmap scan report for mytimes.taylors.edu.my (43.216.68.104)
Host is up (0.0027s latency).
Other addresses for mytimes.taylors.edu.my (not scanned): 43.216.84.70 43.217.0.219
rDNS record for 43.216.68.104: ec2-43-216-68-104.ap-southeast-5.compute.amazonaws.com
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
443/tcp   open  https

Nmap done: 1 IP address (1 host up) scanned in 11.10 seconds

(kaiern@carrot)-[~]
$ nmap -sV mytimes.taylors.edu.my
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-08 10:24 +08
Nmap scan report for mytimes.taylors.edu.my (43.217.0.219)
Host is up (0.0026s latency).
Other addresses for mytimes.taylors.edu.my (not scanned): 43.216.68.104 43.216.84.70
rDNS record for 43.217.0.219: ec2-43-217-0-219.ap-southeast-5.compute.amazonaws.com
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE  VERSION
80/tcp    open  tcpwrapped
443/tcp   open  tcpwrapped

Service detection performed. Please report any incorrect results at https://nmap.org/submi
t/ .
Nmap done: 1 IP address (1 host up) scanned in 24.42 seconds

```

Figure 2 Scanning Results using Nmap

In addition, UDP scans (-sU) showed the states of open | filtered on the UDP port 53 (DNS) and port 123 (NTP). Also, these are standard cloud infrastructure ports rather than application specific services. As a result, their presence does not represent a direct attack surface for ransomware, as they are accessible for user interaction and are largely controlled by the hosting provider.

```

(kaiern@carrot)-[~]
$ sudo nmap -sU -p 53,123 campuscentral.taylors.edu.my
[sudo] password for kaiern:
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-08 10:29 +08
Nmap scan report for campuscentral.taylors.edu.my (103.145.155.69)
Host is up (0.00098s latency).

PORT      STATE      SERVICE
53/udp    open|filtered domain
123/udp   open|filtered ntp

Nmap done: 1 IP address (1 host up) scanned in 7.89 seconds

```

Figure 3 UDP Scanning (campuscentral.taylors.edu.my)

```

(kaiern@carrot)-[~]
$ sudo nmap -sU -p 53,123 taylorslibrary.taylors.edu.my
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-08 10:31 +08
Nmap scan report for taylorslibrary.taylors.edu.my (103.145.154.134)
Host is up (0.00031s latency).

PORT      STATE      SERVICE
53/udp    open|filtered domain
123/udp   open|filtered ntp

Nmap done: 1 IP address (1 host up) scanned in 7.91 seconds

```

Figure 4 UDP Scanning (taylorslibrary.taylors.edu.my)

Table 1 Several Port Scanning Results Table

Domain	Port	Protocol	Service	Status	Notes
university.taylors.edu.my	80	TCP	http – tcpwrapped	Open	Normal for website, fingerprints prevented
	443	TCP	https – tcpwrapped	Open	HTTPS via cloud security layer
mytimes.taylors.edu.my	53	UDP	Domain	Open filtered	Cloud infrastructure DNS behavior
	123	UDP	ntp	Open filtered	Cloud NTP time sync; not application-facing
campuscentral.taylor.edu.my	80	TCP	http – tcpwrapped	Open	Hosted on Microsoft cloud/ SharePoint
	123	UDP	ntp	Open filtered	Cloud time sync used by provider
taylorslibrary.taylor.edu.my	443	TCP	https – tcpwrapped	Open	Standard library web portal
	53	UDP	Domain	Open filtered	Cloud DNS traffic

5.3 MITRE ATT&CK Correlation

Importantly, what showcased during the port scanning is that there are no high-risk ports commonly exploited by ransomware operators were found. For example, ports like SSH/22, which is frequently targeted for brute-force (MITRE ATT&CK, 2018), RDP/3389, that is often exploited for remote access compromise (MITRE ATT&CK, 2020), and SMB/445, a known channel for lateral movement during ransomware campaigns (MITRE ATT&CK, 2020), it is a good practice that none of them are being exposed. As this significantly reduces the likelihood of external ransomware infiltration through network ports. Below figure indicates that the ports that are vulnerable for cyberthreats, especially could act as the ransomware entry point are closed, which is a good safeguard for a enterprise.

```

(kaiern@carrot)-[~]
$ nmap -sV -p 22,3389,445 university.taylors.edu.my
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-08 14:53 +08
Nmap scan report for university.taylors.edu.my (18.161.180.50)
Host is up (0.00014s latency).
Other addresses for university.taylors.edu.my (not scanned): 18.161.180.7 18.161.180.4 18.161.180.82
00:1e:af6b:5980:93a1 2600:9000:2666:5600:1e:af6b:5980:93a1 2600:9000:2666:4e00:1e:af6b:5980:93a1 260
e:af6b:5980:93a1 2600:9000:2666:e800:1e:af6b:5980:93a1 2600:9000:2666:5a00:1e:af6b:5980:93a1
rDNS record for 18.161.180.50: server-18-161-180-50.kul50.r.cloudfront.net

PORT      STATE    SERVICE    VERSION
22/tcp    filtered ssh
445/tcp    filtered microsoft-ds
3389/tcp   filtered ms-wbt-server

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.97 seconds

```

Figure 5 Vulnerable Ports Scanning (port 22,445,3389)

In short, according to the MITRE ATT&CK framework, the primary threat to these systems would fall under T1190, exploit public-facing application (MITRE ATT&CK, 2018), which only becomes relevant if the web application itself contains a vulnerability. In this case, since only the standard web ports of Taylor’s University are exposed and protected by cloud infrastructure simultaneously, the main attack vector remains user-driven threats such as phishing (T1566) rather than direct network exploitation (MITRE ATT&CK, 2020).

Table 2 MITRE ATT&CK Correlation Table

Service/ Vector	MITRE Technique	How attackers use in Ransomware attacks	Relevance to Taylor’s
HTTP (80)	T1190 – Exploit Public-Facing Application	Exploit vulnerable web apps and gain initial access to upload ransomware payloads	Since only standard web ports exposed; risk depends on website security
HTTPS (443)	T1190 – Exploit Public-Facing Application	Same as above but over encrypted traffic where allows hidden delivery of malware if app is vulnerable	Cloud protection reduces exposure
Email/ User Interaction	T1566 – Phishing	Trick users into clicking links or giving credentials, where delivers malware or steals password	Most likely ransomware entry point, very high risk due to large user base
Credentials	T1110 – Brute Force	Guess passwords repeatedly, deploys ransomware from inside once logged in successful	Not exposed externally, but still relevant internally
RDP (3389)	T1021.001 – Remote Desktop Protocol	Gain access through exposed remote desktop and able to run ransomware manually	Port not exposed, low external risk
SMB (445)	T1021.002 – SMB Lateral Movement	Move across internal network which ransomware spreads laterally, encrypting multiple systems	Not exposed externally but important inside the network

Malicious Files/ Links	T1204 – user Execution	Directly executes ransomware or downloader scripts when users open malicious documents	High risk which is very common in education
Cloud Apps (Office365, LMS)	T1078 – valid Accounts	Use stolen credentials to log in and deploys ransomware from inside	High risk if MFA not fully enforced

Overall, all the techniques listed above are part of the initial access, execution, and lateral movement stages used by ransomware groups. While Taylor’s University does not expose high-risk ports such as RDP 3389 or SMB 445 externally, attackers can still enter through phishing, malicious file execution, or stolen valid accounts, which remain the dominant ransomware vector in higher education environments.

6.0 Conclusion

Ransomware attacks have recently become one of the most dangerous threats in the field of cybersecurity for institutions of higher learning around the world, with estimated annual losses likely to be above \$250 billion by 2031. This study offers a systematic risk analysis of the ransomware threat environment faced by Taylor's University. As one of the largest private universities in Malaysia with more than 30,000 students, Taylor's makes a high-stakes target for hackers because of its large number of users, open architecture design for networking purposes, and precious data.

In short, this study used the analysis of the X.800 Security Architecture Framework and the Cybersecurity Framework of NIST in identifying several key security weaknesses: insufficient encryption protection for backup environments and endpoints, insufficient tamper-resistance for log-related mechanisms, and insufficient multi-factor authentication coverage for access to critical university systems. Port scans had shown that Taylor's public-facing networks have only HTTP (port 80) and HTTPS (443) ports open, with risky ports properly shut down, showing that Taylor's has a strong foundation for basic external protection. But when mapping back to MITRE ATT&CK techniques, one sees that a major source of threats comes from user-accessible attack methods such as phishing (T1566) and attacks on public-facing applications (T1190), reaffirming that cybersecurity requires not only technical controls but also institution-wide protection involving people and processes.

The eight-step model for a ransomware risk management process, comprising scope definition, asset inventory, criticality classification, vulnerability identification, risk assessment, risk treatment, control implementation, and continuous monitoring, as outlined above, offers a structured and repeatable model for risk management processes to take place in Taylor's University. The integrated approach in relation to People, Process, and Technology, which has been coupled with useful strategies in countering risks associated with ransomware, has been aligned with corresponding risk points that exist. These relate to either People, where security awareness activities, as well as simulation exercises for phishing attempts, can be used to mitigate risks related to people; Process, which involves developing effective response strategies with robust 3-2-1 approaches for backups that facilitate rapid recovery; and Technology, which adopts strategies such as supplementation of multi-authentication strategies, installing Endpoint Detection and Response technology, and having immutable backups.

A successful attack by a ransomware would result in adverse challenges being faced by Taylor's in many different areas. On one hand, there would be issues pertaining to operations. The disruption of myTImeS, Student Information Systems, test systems, and finance systems would result in complications in both study timetables and graduation timetables. Reputation would also take a hit due to security incidents. In cases related to data confidentiality, “double-extortion” attacks would result in data being encrypted, as well as a potential expose of confidential data. This would also be a violation of Malaysia’s Personal Data Protection Act (PDPA) and result in legal fallout. There would also be potential monetary losses in terms of investigating costs for incidents, potential costs for taking legal advice, and lost productivity, which in total would amount to millions of ringgit.

The issue of cybersecurity needs constant effort and emphasis on surveillance, review, and improvement. This is due to the ever-changing means of launching attacks over a period of time. Therefore, for Taylor's to develop a dynamic security environment that involves constant vulnerability scanning, reviewing of firewall logs, as well as periodic updates of their security policies, it will require that emphasis and allocation of budgets, as well as involvement of all personnel, in addition to use of advanced technologies, be employed to secure Taylor’s key data assets.

This assessment, formulated in line with best practices and models that concentrate on Taylor’s own circumstances, provides opportunities for analysis and recommendations. Implementing this approach would help Taylor’s University develop a robust cyber defense mechanism that would instill their confidence in carrying out their mission of excellence in learning in this contemporary world.

Video link

<https://youtu.be/ab1Zmn69mp0?si=HheyI0sf5EmLocvu>

References

- MITRE ATT&CK. (n.d.). *Remote Services: Remote Desktop Protocol, Sub-technique T1021.001 - Enterprise* | MITRE ATT&CK®. Attack.mitre.org.
<https://attack.mitre.org/techniques/T1021/001/>
- MITRE ATT&CK. (2018). *Chaos, Software S0220* | MITRE ATT&CK®. Mitre.org.
<https://attack.mitre.org/software/S0220/>
- MITRE ATT&CK. (2018). *Exploit Public-Facing Application, Technique T1190 - Enterprise* | MITRE ATT&CK™. Attack.mitre.org. <https://attack.mitre.org/techniques/T1190/>
- MITRE ATT&CK. (2020, February 11). *Remote Services: SMB/Windows Admin Shares, Sub-technique T1021.002 - Enterprise* | MITRE ATT&CK®. Attack.mitre.org.
<https://attack.mitre.org/techniques/T1021/002/>
- MITRE ATT&CK. (2020, March 2). *Phishing, Technique T1566 - Enterprise* | MITRE ATT&CK®. Attack.mitre.org. <https://attack.mitre.org/techniques/T1566/>
- MITRE ATT&CK. (2018c, April 18). *User Execution, Technique T1204 - Enterprise* | MITRE ATT&CK®. Attack.mitre.org. <https://attack.mitre.org/techniques/T1204/>
- MITRE ATT&CK. (2017, May 31). *Valid Accounts, Technique T1078 - Enterprise* | MITRE ATT&CK®. Attack.mitre.org. <https://attack.mitre.org/techniques/T1078/>
- Cichonski, P., Millar, T., Grance, T., & Scarfone, K. (2012). Computer security incident handling guide. *Computer Security Incident Handling Guide*, 2(2).
<https://doi.org/10.6028/nist.sp.800-61r2>
- Eko Yon Handri, Adi, P., & Dana Indra Sensuse. (2023). *Evaluating the People, Process, and Technology Priorities for NIST Cybersecurity Framework Implementation in E-Government*.
<https://doi.org/10.1109/icocics58778.2023.10277024>
- Javaid, M. I., & Iqbal, M. M. W. (2017, April 1). *A comprehensive people, process and technology (PPT) application model for Information Systems (IS) risk management in*

small/medium enterprises (SME). IEEE Xplore.

<https://doi.org/10.1109/COMTECH.2017.8065754>

Moore, M. (2024). *Top Cybersecurity Threats to Watch in 2025*. University of San Diego.

<https://onlinedegrees.sandiego.edu/top-cyber-security-threats/>

Prodan, M., Prodan, A., & Purcarea, A. A. (2015). Three New Dimensions to People, Process, Technology Improvement Model. *New Contributions in Information Systems and Technologies*, 481–490. https://doi.org/10.1007/978-3-319-16486-1_47

Amos, B. (2019). The impact of ransomware on organizations. *Journal of Cybersecurity Research*, 15(3), 234–251. <https://doi.org/10.1016/j.jcsr.2019.03.005>

Cisco Talos. (2023). *Ransomware: A persistent threat in higher education*. Cisco Systems. Retrieved from <https://www.talosintelligence.com/>

Cybersecurity Ventures. (2023). Global ransomware damage costs predicted to reach \$250 billion USD by 2031. Retrieved from <https://cybersecurityventures.com/global-ransomware-damage-costs-predicted-to-reach-250-billion-usd-by-2031/>

FBI Cyber Division. (2023). *Internet crime complaint center (IC3) 2022 annual report*. Federal Bureau of Investigation. Retrieved from <https://www.ic3.gov/>

International Telecommunication Union. (2023). *ITU-T X.800: Security architecture for Open Systems Interconnection for CCITT applications*. ITU-T. Retrieved from <https://www.itu.int/>

Kaspersky Labs. (2023). *Enterprise threats in 2023: Media blackmail, fake data leaks, and more attacks via cloud*. In *Kaspersky Security Bulletin*. Retrieved from <https://www.kaspersky.com/about/press-releases/enterprise-threats-in-2023-media-blackmail-fake-data-leaks-and-more-attacks-via-clouds>

Mandiant. (2023). *Ransomware-as-a-service: Economics and impact*. Mandiant Threat Intelligence. Retrieved from <https://www.mandiant.com/>

National Institute of Standards and Technology. (2018). *Framework for improving critical infrastructure cybersecurity, version 1.1*. U.S. Department of Commerce. Retrieved from <https://www.nist.gov/cyberframework>

National Institute of Standards and Technology. (2020). *Guidelines for managing organizational cybersecurity risks: NIST special publication 800-39*. U.S. Department of Commerce. Retrieved from <https://csrc.nist.gov/>

Ponemon Institute. (2023). *Cost of a data breach report 2023*. IBM & Ponemon Institute. Retrieved from <https://www.ibm.com/reports/data-breach>

Ransomware Task Force. (2021). *Combating ransomware: A comprehensive framework for action*. Institute for Security and Technology. Retrieved from <https://securityandtechnology.org/report/ransomware/>

Sophos. (2023). *State of ransomware 2023*. Sophos Group. Retrieved from <https://www.sophos.com/>

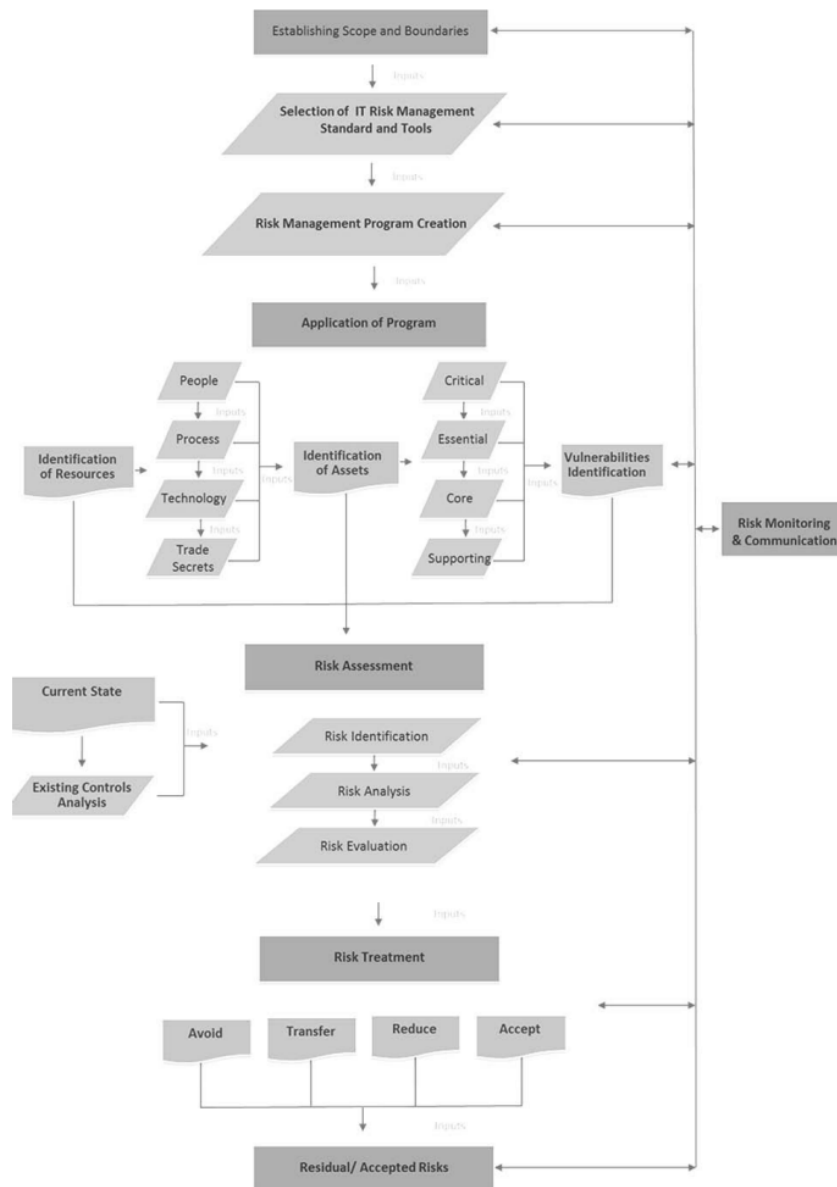
Verizon Communications. (2023). *Data breach investigations report 2023*. Verizon. Retrieved from <https://www.verizon.com/business/resources/reports/dbir/>

CEE Multi-Country News Center. (2024, November 29). *Microsoft Digital Defense Report: 600 million cyberattacks per day around the globe - CEE Multi-Country News Center*. CEE Multi-Country News Center. <https://news.microsoft.com/en-ccc/2024/11/29/microsoft-digital-defense-report-600-million-cyberattacks-per-day-around-the-globe/>

Appendix

PPT Risk Management Application Model Flow from

<https://doi.org/10.1109/COMTECH.2017.8065754>



Assignment Final Report Assessment Criteria

The Assessment of this final assignment report is based on the following criteria:

Group Assessment

Assignment Final Report and Presentation Slides Assessment Rubrics (100 Marks)

Report Content Criteria	Overall Weight (%)	Excellent (10)	Good (8)	Satisfactory (6)	Average (4)	Poor (2)	Marks (M)
Abstract	10	Impactful Abstract	Good Abstract	Average Abstract	Fair Abstract	Poor Abstract	
Background	10	Interesting and Factual Verified Background Writeup	Good description/ explanatory of background	Adequate description/ explanatory of background	Brief description/ explanatory of background	Vague description/ explanation of the background	
		Excellent (25-20)	Good (19-15)	Satisfactory (14-10)	Average (9-5)	Poor (4-0)	

Discussion on the impact of Ransomware to business environment of a company	25	Excellent understanding of the security issues. A great amount of research on the technical issues that caused the security issues.	Good understanding of the security issues. A good amount of research on the technical issues that caused the security issues.	Adequate understanding of the security issues. Some amount of research on the technical issues that caused the security issues.	Lack of understanding of the security issues and lack of research on the technical issues caused the security issues.	Unable to provide a clear understanding of the security issues.	
Analysis, approaches and solutions to address Ransomware in your organisation	25	Able to provide a very clear and reasonable evaluation with detail explanation of its solutions to organisations	Able to provide a good and reasonable evaluation with detail explanations of the solutions proposed to organisations.	Able to provide a reasonable evaluation with fair explanations of the solutions proposed to organisations.	Able to provide an average evaluation with some explanations of the solutions proposed to organisations.	Poorly provided evaluation and lack explanations of the solutions proposed to organisations.	
		Excellent (20 -15)	Good (14-10)	Satisfactory (9-5)	Average (4)	Poor (1)	
Clarity and Creativity of Presentation Video	20	Excellent Clarity with High Creativity Presentation (Strategically and Technically Effective)	Good Clarity and Creativity Presentation (Strategically and technically Good)	Fair Clarity and Creativity Presentation (Strategically and technically Fair)	Acceptable Clarity and Creativity Presentation (Strategically and technically adequate)	Lack of clarity and low creativity presentation (Strategically and technically ineffective)	
		Excellent (5)	Good (4)	Satisfactory (3)	Average (2)	Poor (1)	

Conclusion	5	Comprehensive and effective coverage	Good coverage	Adequate coverage	Average coverage	Poor coverage	
Quality References	5	25 or more of recent quality references	20-25 recent quality references	15 -20 recent quality references	10-15 recent quality references	Less than 10 recent quality references	
Total:	100						